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Computer Programming using Java-II PRACTICE BOOK (SEM-II-EVEN_2025-CE RELATED BRANCHES)

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
UNIT 1- Abstraction & Interface with Runtime Polymorphism								
TOPIC NAME:- Dynamic method dispatch, Object casting and instance of operator (MCQS)								

1	1	Output of following Java program?
		class Base {
		public void Print() {
		System.out.println("Base");
		}
		}
		class Derived extends Base {
		public void Print() {
		System.out.println("Derived"); }
		}
		class Main{
		public static void DoPrint(Base o) {
		o.Print(); }
		public static void main(String[] args) {
		Base x = new Base();
		Base y = new Derived();
		Derived z = new Derived();
		DoPrint(x);
		DoPrint(y);
		DoPrint(z);
		} }
			A	1	Base
					Derived
					Derived
						Base
						Base
						Derived
							Base
							Derived
							Base
								Compiler Error

2	1	class Base {
		public void Print() { System.out.println("Base");
		} }
		class Derived extends Base { public void Print() {
		System.out.println("Derived");
		} }
		class Main{
		public static void DoPrint(Derived o) { o.Print(); }
		public static void main(String[] args) { Base x = new Base();
		Base y = new Derived(); Derived z = new Derived();
		DoPrint((Derived)x);
		DoPrint((Derived)y);
		DoPrint(z); }}
			C	1	Base
					Derived
					Derived
						Base
						Base
						Derived
							Runtime Error	Compiler Error

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3	1	Will this code run successfully? If yes, what will be the output?
		class A {
		private int x = 50;
		void m1() {
		System.out.println(x);
		}
		}
		class B extends A {
		}
		public class Test {
		public static void main(String[] args) {
		A a = new B();
		a.m1();
		} }
			C	1	30	40	50	Compiler error

4	1	Predict the output of following program. Note that foo() is public in base and private in derived.
		class Base {
		public void foo() { System.out.println("Base"); }
		}
		class Derived extends Base {
		private void foo() { System.out.println("Derived"); }
		}
		public class Main {
		public static void main(String args[]) {
		Base b = new Derived();
		b.foo();
		}
		}
			C	1	Base	Derived	Compiler Error	Runtime Error

5	1	What will be the output of the following code?
		public class A {
		int x = 20;
		}
		class B extends A {
		int x = 30;
		}
		class Test {
		public static void main(String[] args) {
		B b = new B();
		System.out.print(b.x+",");
		A a = new A();
		System.out.print(a.x+",");
		A a2 = new B();
		System.out.print(a2.x+","); } }
			B	1	20,20,30,	30,20,20,	30,30,30,	30,20,30,

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6	1	class Base {
		public static void show() {
		System.out.println("Base::show() called");
		}
		}
		class Derived extends Base {
		public static void show() {
		System.out.println("Derived::show() called");
		} }
		class Main {
		public static void main(String[] args) {
		Base b = new Derived();
		b.show();
		} }
			A	1	Base::show() called	Derived::show() called	Compiler Error	None of these

7	1	class A {
		void m1(String x){
		System.out.print("One ");
		} }
		class B extends A {
		public void m1(String x){
		System.out.print("Two ");
		super.m1(null);
		} }
		class Test{
		public static void main(String[] args){
		A a = new B();
		a.m1(null);
		} }
			B	1	One Two	Two One	null null	Compiler error

TOPIC NAME:- Dynamic method dispatch, Object casting and instance of operator (PROGRAMS)								
8	1	Write a program that implementts dynamic method dispatch.		4				
9	1	Create class Fruit with method void display(). Create subclasses Apple and Banana that overrides display(). In main method kindly show the concept of Dynamic method dispatch by creating reference of Fruit and object of Apple and Banana respectively and call display().		4				
10	1	Declare a class called book having author_name as private data member. Extend book class to have two sub classes called book_publication&paper_publication. Each of these classes have private member called title. Write a complete program to show usage of dynamic method dispatch (dynamic polymorphism) to display book or paper publications of given author. Use command line arguments for inputting data.		5				
TOPIC NAME:- Abstract class, Abstract Method (MCQS)								
11	1	Which of these keywords are used to define an abstract class?	B	1	abst	abstract	Abstract	abstract class

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
12	1	What is the correct statement in java abstract class? A) In abstract class every method must be abstract B) In Abstract class we have to define atlease one abstract method. C) In abstract class if we do not define any abstract method then it will not give any error. D) In abstract class every method is public, static and abstract	D	1	A & B	only D	only A	Only C

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

13	1	What is the output of the following code?
		abstract class LJU
		{
		LJU() {
		System.out.print("Hi ");
		}
		}
		class LJIET extends LJU
		{
		LJIET() {
		System.out.print("Bye");
		}
		}
		class AbstractClassTesting
		{
		public static void main(String[] args)
		{
		LJU lj = new LJIET();
		}
		}
			B	1	Hi	Hi Bye	Bye	Bye Hi

14	1	What is the output of the following code?
		abstract class Book
		{
		abstract void show();
		}
		abstract class Fiction extends Book
		{
		abstract void print();
		}
		class BookStore extends Fiction
		{
		void show()
		{
		System.out.println("Book Store Show");
		}
		void print()
		{
		System.out.println("Book Store Print");
		}
		}
		class AbstractClassTesting4
		{
		public static void main(String[] args)
		{
		BookStore bs = new Fiction();
		bs.show(); } }
			C	1	Book Store Show	No Output	Compile time error	Run time error

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

15	1	What will be the output of the following code?
		abstract class Book {
		abstract void show(); }
		abstract class Fiction extends Book { abstract void print();
		void show() {
		System.out.println("Book Store Show");
		}
		}
		class BookStore extends Fiction {
		void show() {
		System.out.println("Book Store Print");
		}
		void print() {
		System.out.println("Book Store Show");
		}
		}
		class AbstractClassTesting4 {
		public static void main(String[] args)
		{
		Fiction bs = new BookStore();
		bs.show();
		} }
			B	1	Book Store Show	Book Store Print	Compile time error	Run time error

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
TOPIC NAME:-Abstract class, Abstract Method (PROGRAMS)								
16	1	Write a abstract class named Person and its two subclasses named student and Employee. A person has a name, address, phone number and email address. A student has enrollment course. An Employee has an office, salary, and designation. Define constructors and methods for input and display for both classes. Define constructor and methods for input and display for both classes. Write a main program to give demonstration of all.		5				
17	1	Describe abstract class called Shape which has three sub classes say Triangle, Rectangle, Circle. Define one method area() in the abstract class and override this area() in these three subclasses to calculate for specific object i.e. area() of Triangle subclass should calculate area of triangle etc. Same for Rectangle and Circle.		5				
18	1	The abstract Vegetable class has three subclasses named Potato, Brinjal and Tomato. Write an application that demonstrates how to establish this class hierarchy. Declare one instance variable of type String that indicates the color of a vegetable. Create and display instances of these objects. Override the toString() method of Object to return a string with the name of the vegetable and its color.		5				
TOPIC NAME:- Interface: Implementation of Interface(Partial & Full) Extend Interface (MCQS)								
19	1	All Interface variables are ___ by default in Java.	C	1	public Only	final Only	public, static and final	None
20	1	All Interface method are ___ by default in Java.	C	1	public	abstact	public and abstact	final
21	1	A Class implementing an Interface can use ____ access modifier before the implemented methods.	A	1	public	private	protected	default
22	1	Can you call it a full-fledged inheritance of using ABSTRACT classes and INTERFACES in Java?	A	1	NO	YES	There is no such concept of absract class in JAVA	None of these

23	1	Which two classes use the Shape class correctly?	B	1	public class Circle implements Shape
					{
					private int radius;
					}
						public abstract class Circle extends Shape
						{
						private int radius;
						}
							public class Circle extends Shape
							{
							private int radius;
							public void draw();
							}
								public abstract class Circle implements Shape
								{
								private int radius;
								public void draw();
								}

24	1	Which is the keyword used to implement interface in Java?	B	1	extends	implements	instanceof	None of these
25	1	Choose a correct statement about Java Interfaces?	D	1	Interface contains only abstract methods by default.	A Java class can implement multiple interfaces	An Interface can extend or inherit another Interface.	All the mention

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

26	1	What will be the output of the following code?
		interface Book
		{
		char type='C';
		}
		public class InterfaceTest3
		{
		public static void main(String[] args)
		{
		System.out.println(new Book().type);
		}
		}
			D	1	C	No Output	run time error	Compiler error

27	1	Which of the following is correct? A: All methods in an interface are by default public. We can not change the access specifier to private or protected. B: Abstract methods have to be either protected or public. They can not be private	B	1	Both are incorrect	Both are Correct	only A	Only B

28	1	What is the output of the following code?
		
		class King {
		public static void main(String[] args) {
		King k = new King();
		Elephant e = k.new Elephant();
		System.out.print(e.step2(2, 3));
		}
		interface Queen {
		float step2(int low, int high);
		}
		interface Pawn {
		float step3(int a, int b, int c);
		}
		abstract class Knight implements Queen, Pawn {
		}
		class Elephant implements Queen {
		public float step2(int x, int y) {
		return 2;
		}
		}
		}
			C	1	compile time error	run time error	2.0	2

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

29	1	class King {
		public static void main(String[] args) {
		King k = new King();
		Elephant e = k.new Elephant();
		System.out.print(e.step2(3, 2));
		}
		interface Queen {
		float step2(int low, int high);
		}
		interface Pawn {
		float step3(int a, int b, int c);
		}
		abstract class Knight implements Queen, Pawn {
		}
		class Elephant implements Queen
		{
		public float step2(int x, int y) {
		return 1;
		}
		}
		}
			C	1	compile time error	run time error	1.0	1

30	1	What will be the output of the following code?
		interface Bus
		{
		void move();
		}
		class ElectricBus implements Bus
		{
		public void move()
		{
		System.out.println("Implemented move() method.");
		}
		}
		public class InterfaceTest1
		{
		public static void main(String[] args)
		{
		new ElectricBus().move();
		}
		}
			A	1	No output	Implemented move() method.	run time error	Compiler error

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

31	1	What will be the output of the following code?
		interface LJU1 {
		void exam();
		}
		abstract class LJIET1 implements LJU1{
		public void exam(){
		System.out.println("LJIET Exam");
		}
		}
		class FY extends LJIET1{
		public void exam(){
		System.out.println("FY Exam");
		}
		}
		class FacT1{
		public static void main(String args[]){
		LJIET1 lj = new FY();
		lj.exam();
		}
		}
			A	1	FY Exam	LJIET Exam	No output	compile time error

32	1	What will be the output of the following code?
		interface X {
		void show();
		}
		interface Y {
		void print();
		}
		interface Z extends X, Y{
		void display();
		}
		abstract class LJFac implements Z{
		public void show(){
		print();
		}
		public void print(){
		System.out.print("-Hi ");
			D	1	Compile time error : display() Not Found	Compile time error: cannot extend LJFac	Hi-Hi	-Hi Hi

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		System.out.print("Hi");
		display();
		}
		}
		class LJFYFac extends LJFac{
		public void display(){
		System.out.print("Hi");
		} }
		class TestEx{
		public static void main(String args[]){
		LJFYFac f = new LJFYFac();
		f.show(); } }						

33	1	What will be the output of following java program?
		interface calculate
		{
		int VAR = 0;
		void cal(int item);
		}
		class display implements calculate {
		int x;
		public void cal(int item)
		{
		if (item<2)
		x = VAR;
		else
		x = item * item;
		} }
		class interfaces {
		public static void main(String args[])
		{
		display[] arr=new display[3];
		for(int i=0;i<3;i++)
		arr[i]=new display();
		arr[0].cal(0);
		arr[1].cal(1);
		arr[2].cal(2);
		System.out.print(arr[0].x+" " + arr[1].x + " " + arr[2].x);
		} }
			C	1	0 1 2	1 1 4	0 0 4	0 1 4

TOPIC NAME:- Interface: Implementation of Interface(Partial & Full) Extend Interface (PROGRAMS)								
34	1	Java program to implement interface with hierarchical inheritance		3				
35	1	Write a java program to explain runtime polymorphism using interfaces		3				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
36	1	Write a program that demonstrates the instanceof operator.Declare interfaces I1 and I2. Interface I3 extends both of these interfaces.Also declare interface I4.class X implements I3.Class W extends X and implements I4.create an object of class W.Use the instanceof operator to test if that object implements each of the interfaces and is of type X.		4				
37	1	Write a java program to implement the multiple inheritance concepts for calculating area of circle and square		4				
38	1	Write a program that illustrates interface inheritance. Interface A is extended by A1 and A2. Interface A12 inherits from both P1 and P2.Each interface declares one constant and one method. Class B implements A12.Instantiate B and invoke each of its methods. Each method displays one of the constants.		4				
39	1	Create interface Univerity with method takeExam(). Create Interface State with method getName(). Create interface StateUniversity that extends University and State. Create print() inside StateUniversity. Create class TestState that implements StateUniversity interface. and Create a class with main method that calls all methods of TestState class		4				
40	1	Write a program that illustrates interface inheritance. Interface A is extended by A1 and A2. Interface A12 inherits from both P1 and P2.Each interface declares one constant and one method. Class B implements A12.Instantiate B and invoke each of its methods. Each method displays one of the constants		4				
41	1	Write a program that illustrates interface inheritance. Interface P is extended by P1 And P2. Interface P12 inherits from both P1 and P2.Each interface declares one constant and one method. Class Q implements P12.Instantiate Q and invokes each of its methods. Each method displays one of the constants		4				
42	1	Write a program to demonstrate the multipath inheritance for the classes having relations as shown in figure A-> (B, C) ->D. A / \ B C \ / D		5				
43	1	Explain single level and multiple inheritances in java. Write a program to demonstrate combination of both types of inheritance as shown in figure 1. i.e.hybrid inheritance . A B \ / C | D		5				

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44	1	The Transport interface declares a deliver() method. The abstract class Animal is the superclass of the Tiger, Camel, Deer and Donkey classes. The Transport interface is implemented by the Camel and Donkey classes. Write a test program that initialize an array of four Animal objects. If the object implements the Transport interface, the deliver() method is invoked.		5				
45	1	Create an interface named Polygon. It has a default method getSides() and an abstract method getArea().create two classes Rectangle and Square that implement Polygon.The Rectangle class provides the implementation of the getArea() method and overrides the getSides() method. the Square class only provides the implementation of the getArea() method. while calling the getSides() method using the Rectangle object, the overridden method is called. However, in the case of the Square object, the default method is called.		5				
46	1	Write a program to demonstrate the multipath inheritance for the classes having relations as shown in figure A B C \ / / D / | / E		5				
47	1	Q1)Create an abstract class pen with methods write () and refill () as abstract methods Q2)Use the pen class from Q1 to create a concrete class fountain pen with additional method change Nib () Q3)Create a class monkey with jump ( ) and bite ( ) methods Create a class human whichinherits this monkey class and implements basicanimal interface with eat ( ) and sleep methods Q4)Create a class telephone with ( ) , lift ( ) and disconnected ( ) methods as abstract methods create another class smart telephone and demonstrate polymorphism Q5)Demonstrate polymorphism using using monkey class from Q3 Q6)Create an interface TVremote and use it to inherit another interface smart TVremote Q7)Create a class TV which implements TVremote interface from Q6		5				

UNIT - 2 - Introduction to Package

TOPIC NAME:- Use of Package, Import statement, Access Modifiers , Access control with example (MCQs)

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
48	2	Which of the following is true about inheritance in Java? 1) Private methods are final. 2) Protected members are accessible within a package and inherited classes outside the package. 3) Protected methods are final. 4) We cannot override private methods.	A	1	1, 2 and 4	Only 1 and 2	1, 2 and 3	2, 3 and 4
49	2	A class member declared protected becomes a member of subclass of which type?	B	1	public Member	Private Member	Protected Member	Static Member
50	2	Which of these packages contains abstract keyword?	A	1	java.lang	java.util	java.io	java.system
51	2	Which of these keywords is used to define packages in Java?	C	1	pkg	Pkg	package	Package
52	2	What is the maximum number of levels or depth up to which sub-packages can be defined in Java?	D	1	8	16	32	There is no limit
53	2	Which declaration statement should be the first statement in a Java file	C	1	class	inteface	package	import
54	2	Which is the symbol used to separate a super-package and a sub-package at the time of declaring or importing in a Java program?	A	1	Period (.) or DOT	Dollar ($)	Pound (#)	Arrow (->)

55	2	Will this code run successfully? If yes, what will be the output?
		public class A
		{
		private int x = 50;
		void m1() {
		System.out.println(x);
		} }
		public class B extends A {
		}
		public class Test {
		public static void main(String[] args)
		{
		A a = new B();
		a.m1();
		} }
			C	0.5	0	50	Compile time error	Run time error

56	2	Which of this access specifiers can be used for a class so that its members can be accessed by a different class in the same package?	D	1	public	protected	default	all of the mention
57	2	Which access specifiers can be used for a class so that it’s members can be accessed by a different class in the different package ?	A	1	public	protected	default	all of the mention
58	2	Choose the correct statement about access modifiers in Java. A) public, protected and default variables and methods can be accessed outside the package somehow. B) public and protected variables and methods can be accessed outside the package somehow. C) Only public variables and methods can be accessed outside the package somehow. D) all of the above	B	1	Only A	Only B	Only C	B and C

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

59	2	Will this code compile successfully? If yes, what is the output? If no, identify the errors.
		
		package pack1;
		public class A
		{
		private int x = 50;
		protected int y = 100;
		int z = 200;
		}
		package pack2;
		import pack1.A;
		public class B extends A {
		
		}
		import pack2.B;
		public class Test {
		public static void main(String[] args)
		{
		B b = new B();
		System.out.println(b.x);
		
		System.out.println(b.y);
		System.out.println(b.z);
		}
		}
			D	1	50	100	200	Compiler Error

60	2	Which of the following is an incorrect statement about packages?	D	1	Package defines a namespace in which classes are stored	A package can contain other package within it	Java uses file system directories to store packages	A package can be renamed without renaming the directory in which the classes are stored
61	2	The name of a package is the name of the ___ in Java.	D	1	folder	All parent folders separated by DOT symbols	All parent packages separated by DOT symbols	All the above
62	2	Which is the correct syntax to import a Java package below? A) import java.package.*; B) import java package1.package2.*; C) import java package1.package2.Class; D) import java package1.package2.Class.*;	A	1	Only A	Only B	Only C	B and C

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
63	2	Choose a correct statement about using the classes or interfaces or abstract classes inside the packages in our Java program. A) You can extend the class imported from the package. B) You can implement the interfaces imported from the package. C) You can extend the abstract classes imported from the package.	D	1	Only A	Only B	B and C	A,B and C

64	2	What is the output of the below Java program with packages?
		
		//MathClas.java
		package package1;
		public class MathClass
		{
		public static int getRandom()
		{
		//returns a random number from 0 to 999
		return ((int) (Math.random() * 1000));
		}
		}
		//PackageTest3.java
		import package1.*;
		public class PackageTest3
		{
		public static void main(String[] args)
		{
		System.out.print(MathClass.getRandom());
		} }
			C	1	No output	Compiler error	484 (some random number)	run time error

65	2	package pack1;
		public class A {
		public int x = 10;
		protected int y = 20;
		private int z = 30;
		}
		
		package pack2;
		import pack1.A;
		public class B extends A {
		public void display() {
		System.out.print(x+ " ");
		System.out.println(y+ " ");
		// System.out.println(z);
		}
			B	1	20 10	10 20	Compile time error	Run time error

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

		}
		//Test.java
		
		import pack2.B;
		public class Test {
		  public static void main(String[] args) {
		  B b = new B();
		  b.display();
		  }
		}
		What will be the output of the code?						

66	2	Please consider following java code :
		
		package university;
		
		public class Student {
		  protected String name;
		  int age;
		  private String studentId;
		
		  public Student(String name, int age, String studentId) {
		  this.name = name;
		  this.age = age;
		  this.studentId = studentId;
		  }
		
		  protected void displayInfo() {
		  System.out.println("Name: " + name);
		  System.out.println("Age: " + age);
		  System.out.println("Student ID: " + studentId);
		  }
		}
		package university;
		
		public class Main {
		  public static void main(String[] args) {
		  Student student = new Student("Ajay ", 20, "12345");
		  student.displayInfo();
		  }
		} What will be the output ?
			A	1	The code will compile successfully and display the following output:
					Name: Ajay
					Age: 20
					Student ID: 12345
						The code will compile successfully but throw a runtime error.
						Explanation: The displayInfo() method is accessible within the Main class due to package-private access. Therefore, the code will compile without errors but display the student information correctly.
							The code will fail to compile due to an error in the displayInfo() method.
							Explanation: The displayInfo() method has package-private access, allowing it to be accessed only within the same package. Since Main class is in a different package, it cannot access this method directly.
								The code will compile successfully but won't display any output.
								Explanation: Although the code compiles successfully, there is no System.out.println() statement in the main method to display the student information.

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

67	2	What is the output of the following code?
		
		class A
		{
		public int i;
		private int j;
		}
		class B extends A
		{
		void display()
		{
		super.j = super.i + 2;
		System.out.println(super.i + " " + super.j);
		}
		}
		class inheritance
		{
		public static void main(String args[])
		{
		B obj = new B();
		obj.i=1;
		obj.j=2;
		obj.display();
		} }
			D	1	2 2	3 3	1 2	compile time error

TOPIC NAME:- Use of Package, Import statement, Access Modifiers , Access control with example (PROGRAMS)								
68	2	Create class p1 in package 1 and that class calculate the reverse of a number (take user input and protected method). Create a class Test in package 2 which print the reverse of number of package 1.		3				
69	2	Write a Program to create user defined package and to calculate marks & percentage of students.		4				
70	2	Create a multiple package and show the uses of different access modifiers.		4				
71	2	Create class circle in package 1 and that class calculate the area of circle. Create a class Test in package 2 which print the area of circle of package 1.		4				
72	2	Create a multiple package which show the difference between protected and default access modifier.		4				
73	2	package P contain a class One that has x as data member. package Q contain a class Two that has y as data member .package R contain a class Test that has add() method which add X and Y of class One and Two.		5				
UNIT 3- Exception handling								
TOPIC NAME:- Types of Errors, checked and unchecked Exception (MCQS)								

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
74	3	What is the base class for all Exception ?	B	1	java.lang.Exception	java.lang.Throwable	java.lang.RuntimeExceptio n	java.lang.Error
75	3	These exceptions are seen in NullPointerException, ArrayIndexOutOfBoundsException, ArithmeticException,NumberFormatException ?	B	1	Checked exception	Unchecked exception	Both A and B	none of the mention
76	3	The built-in base class in Java, which is used to handle all exceptions is	C	1	Raise	Exception	Throwable	Error

77	3	Predict the output of following Java program
		class A{
		public static void main(String[] args) {
		int[] a = new int[5];
		a[10] = 50;
		System.out.println(a[3]);
		} }
			A	1	ArrayIndexOutOfBoundsEx ception	NullPointerException	ArithmaticException	NumberFormateException

TOPIC NAME:- Exception handling mechanism: Use of try, catch, throw, throws and finally (MCQS)								
78	3	Which of these keywords is not a part of exception handling?	D	1	try	catch	finally	thrown

79	3	class p{
		public static void main(String[] args) {
		try {
		System.out.println("Hello");
		}
		}
		}
			A	1	compile time error	Run time error	Both A and B	None of the mention

80	3	Which of these keywords must be used to handle the exception thrown by try block in some rational manner?	C	1	try	finally	catch	thrown
81	3	Which of these keywords is used to manually throw an exception?	D	1	try	finally	catch	throw
82	3	Which of the following statements is true about exception handling in Java:	A	1	A try block can have many catch blocks but only one finally block	A try block can have many catch blocks and many finally blocks	A try block must have one finally block for each catch block	A try block must have at least one catch block to have a finally block
83	3	Which keyword is used to handover our created exception objects to the JVM manually?	B	1	try	throw	throws	break

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

84	3	What will be the output of the following Java program?
		class exception_handling
		{
		public static void main(String args[])
		{
		try
		{
		System.out.print("Hello" + " " + 1 / 0);
		}
		catch(ArithmeticException e)
		{
		System.out.print("World");
		}
		}
		}
			B	1	Hello	World	HelloWorld	Hello World

85	3	What will be the output of the following Java program?
		class exception_handling {
		public static void main(String args[]) {
		try {
		int a, b;
		b = 0;
		a = 5 / b;
		System.out.print("A"); }
		catch(ArithmeticException e)
		{
		System.out.print("B");
		}
		finally
		{
		System.out.print("C");
		}
		}
		}
			D	1	A	B	AC	BC

86	3	Predict the output of following Java program
		class Main {
		public static void main(String args[]) {
		try {
		throw 10;
		}
		catch(int e) {
		System.out.println("Got the Exception " + e);
		}
		}
		}
			C	1	Got the Exception 10	Got the Exception 0	Compile time error	run time error

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

87	3	Predict the output of following Java program
		class Hello{
		public static void main(String[] args) {
		System.out.print("1");
		try{
		System.out.print("2");
		}
		finally{
		System.out.println("3");
		}
		}
		}
			A	1	123	1
						2
						3
							Compile time error	Run time error

88	3	What will be the output of the following Java program?
		class P2 {
		public static void main(String[] args) {
		try {
		System.out.println("FY1");
		}
		finally {
		System.out.println("FY2");
		}
		catch (Exception e) {
		System.out.println("FY3");
		}
		}}
			C	1	FY1FY2FY3	FY1FY3FY2	compile time error	run time error

89	3	What will be the output of the following code?
		class P3 {
		public static void main(String[] args) {
		try {
		System.out.print(1 + ",");
		} catch (ArithmeticException e) {
		System.out.print(7 + ",");
		} finally {
		System.out.print(8 + ",");
		}
		System.out.print(9 + ",");
		}
		}
			D	1	1,9,8,	1,7,8,	7,8,9,	1,8,9,

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

90	3	Predict the output of following Java program
		class TestException {
		static int a,b;
		public static void main(String args[]) {
		try{
		int a1 = Integer.parseInt(args[2]);
		int b1 = Integer.parseInt(args[1]);
		int c = a1+b1/(a+b);
		}
		catch(ArrayIndexOutOfBoundsException e) {
		System.out.println("Hello"); }
		catch(ArithmeticException e) {
		System.out.println("Bye");
		}
		catch(Exception e) {
		System.out.println("LJU");
		} } }
		if I run java code such as : java TestException 0 2 3
		What will be the output?
			B	1	LJU	Bye	Compile time error	run time error

91	3	Predict the output of following Java program
		class Test
		{
		public static void main(String[] args) {
		try
		{
		int a[]= {1, 2, 3, 4};
		for (int i = 1; i <= 4; i++) {
		System.out.println ("a[" + i + "]=" + a[i] + "n");
		}
		}
		catch (Exception e) {
		System.out.println ("error = " + e);
		}
		catch (ArrayIndexOutOfBoundsException e)
		{
		System.out.println ("ArrayIndexOutOfBoundsException");
		}
		} }
			D	1	ArrayIndexOutOfBoundsException	Error Code is printed	some other output	compile time error

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

92	3	What will be the output of following java program?
		class Test {
		public static void main(String[] args) {
		String s="null";
		try {
		System.out.print("n");
		System.out.print("u");
		System.out.print("l");
		System.out.print(s.length());
		return;}
		catch(NullPointerException e) {
		System.out.print("NPE"); }
		finally {
		System.out.print("l"); }
		System.out.print("null"); } }
			D	1	nulNPElnull	nulNPE4	nullnull	nul4l

93	3	What will be the output of the following code If I Run it directly ?
		class TestException {
		public static void main(String args[])
		{ int a,b;
		try{
		
		int a1 = Integer.parseInt(args[2]);
		int b1 = Integer.parseInt(args[1]);
		int c = a1+b1/(a+b);
		}
		catch(ArrayIndexOutOfBoundsException e) {
		System.out.println("Hello");
		} catch(ArithmeticException e)
		{ System.out.println("Bye"); }
		catch(Exception e) {
		System.out.println("LJU");
		}
		}
		}
			A	1	Hello	LJU	Bye	Run time Error : java.lang.ArrayIndexOutOf BoundsException : Index 1 Out of Bound

TOPIC NAME:- Exception handling mechanism: Use of try, catch, throw, throws and finally (PROGRAMS)								
94	3	Write a program to handle ArrayIndexOutofBoundsException using try-catch-finally and throw.		3				
95	3	Write a program to handle NullPointerException.		4				
96	3	Write an application that searches through its command-line argument. If first command line argument is found zero then display error message.		4				
97	3	Write an application that searches through its command-line argument. If sum of first and second command line argument is found 10 then display error message.		4				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
98	3	Write an application that searches through its command-line argument. If any of command line argument is found nagative then display error message.		5				
TOPIC NAME:- Built in Exception, Custom Exception (MCQS)								
99	3	Which are the child class of Throwable class?	C	1	Exception	Error	Both A and B	None of the mention

100	3	Predict the output of following Java program
		class Base extends Exception {}
		class Derived extends Base {}
		public class Main {
		public static void main(String args[]) {
		// some other stuff
		try {
		// Some monitored code
		throw new Derived();
		}
		catch(Base b) {
		System.out.println("Caught base class exception");
		}
		catch(Derived d) {
		System.out.println("Caught derived class exception");
		}
		}
		}
			D	1	Caught base class exception	Caught derived class exception	Compiler Error because derived is not throwable	Compiler Error because base class exception is caught before derived class

101	3	What will be the output of following java program?
		class Test {
		public static void main(String[] args) {
		try
		{
		ArithmeticException e= new ArithmeticException("Division ");
		throw e;
		}
		catch(Exception e)
		{
		System.out.print(e.getMessage());
		}
		System.out.println("/0");
		} }
			D	1	Division	/0	Division/0	Division /0

TOPIC NAME:- Built in Exception, Custom Exception (PROGRAMS)								
102	3	Write a program to handle InterruptedException, IllegalArgumentException using try-cat-finally and throw		3				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
103	3	Write a method for computing x ^ y by doing repetitive multiplication. x and y are of type integer and are to be given as command line arguments. Raise and handle exception(s) for invalid values of x and y. Also define method main.		4				
104	3	Write an application that searches through its command-line argument. If an argument is found that does not begin with an upper case letter, display error message and terminate.		4				
105	3	Write a program to create user define exception MyException. Define a class ExceptionDemo, that has a method named compute(int a) which throws a MyException object, when compute(int a)'s integer parameter is greater than 10		4				
106	3	Write a program to create user define exception MyException. Define a class ExceptionDemo that has a method named compute(int a) which throws a MyException object, when compute(int a)'s integer parameter is divisible by 7 and not divisible by 5.		4				
107	3	Write a program to enter the five subject marks out of 100 of student. If any subject marks is less than 35 then programm will generate exception		4				
108	3	You are going to predict rain in today's exam. Ask user to enter Temperature (in Celsius) , Humidity (in percentag), and Wind Speed(km/h) Throw an Exception : RainException() : if temperature is >=25 , Humidity is > = 70 and Wind Speed > = 15 & Handle it to notify user that "Please keep Umbrella with you" Otherwise Throw LessRainException() : & Handle it to notify user that "Less Possibility of Rain You Can Enjoy Your Day" Note: Both must be Unchecked Exceptions		3				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
109	3	It is required to maintain and process the status of total 9 resources. The status value is to be stored in an integer array of dimension 3x3. The valid status of a resource can be one of the 2 followings: free: indicated by integer value 0 occupied: indicated by integer value 1 inaccessible: indicated by integer value 2 Declare a class called ResourcesStatus, having data member called statusRef, referring to a two dimensional array (3x3) of integers to be used to refer to the above mentioned status values. Define a member method called processStausCount that counts and displays total number of free resources, total number of occupied resources and total number of inaccessible resources. The exception to be raised and handled if total number of occupied resources exceeds total number of free resources. The handler marks status of all inaccessible resources as free. Accept initial status values from command line arguments and initialize the array. Raise and handle user defined exception if invalid status value given		5				
110	3	Write a complete program to accept N integer numbers from the command line. Raise and handle exceptions for following cases : - when a number is –ve - when a number is evenly divisible by 10 - when a number is greater than 1000 and less than 2000 - when a number is greater than 7000 Skip the number if an exception is raised for it, otherwise add it to find total sum		5				
111	3	Declare a class called coordinate to represent 3 dimensional Cartesian coordinates(x, y and z). Define following methods: - constructor - display, to print values of members - add_coordinates, to add three such coordinate objects to produce a resultant coordinate object. Generate and handle exception if x, y and z coordinates of the result are zero. - main, to show use of above methods.		5				

TOPIC NAME:- 9 marks (PROGRAMS)

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
112	1,2,3	In LJ Innovation 2024, We have number of projects and from it we selected project “SmartHub” for incubation center. Their idea is to make smart home automation devices and sell out to the end users. For this they require a Java Software so Student, please help them to automate their smart home using Java. All the best for it. Requirements: Device Interface : This defines behavior of all devices. Methods are : -void turnOn(): turns the device on , -void turnoff(): turns the device off , -boolean getStatus() : returns the status of device whether it is on/off Abstract class : HomeAutomation This provide functionality to manage the devices. -abstract void addDevice(Device device): Adds a new device to the system. -abstract void removeDevice(Device device): Removes a device from the system. -abstract void checkDeviceStatus(Device device) throws DeviceNotFoundException: Checks the status of a specific device. -Create DeviceNotFoundException : If user enters the device which is not there then it throws DeviceNotFoundException() [This is unchecked exception] -abstract void turnOffAllDevices(): Turns off all devices in the system. Light class implements Device: Methods: -void turnOn() : Turns the light On -void turnOff() : Turns the light Off -boolean getStatus(): Returns the current status of the light SecurityCamera class implements Device: Methods: -void turnOn() : Activates the Security Camera -void turnOff() : Deactivates the Security Camera -boolean getStatus(): Returns the current status of the Security Camera Class SmartHub This class contains the main() method and extends the HomeAutomation class. Implement all abstract methods of HomeAutomation class over here in SmartHub class. Create a method handleUserChoice(int choice) : that throws InvaildChoiceException		9				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
		InvalidChoiceException This is must be a checked exception. Ask user to do the following again and again until user enters 8 : -Choice 1 : to Add Light -Choice 2 : to Add SecurityCamera -Choice 3 : To turn Light On [If it is off] / Off [If it is On] -Choice 4 : To turn Security Camera Activate [If it is Deactivate] / Deactivate[If it is Activate] -Chocie 5 : To Check Status of Light : Throw DeviceNotFoundException if Light is not added in Home -Choice 6: TO Check Status of SecurityCamera : Throw DeviceNotFoundException & Handle it if SecurityCamera is not added in Home -Choice 7 : To TurnOffAll Devices -Choice 8 : To Come out from the Choices -If user enter < 1 && >8 then throw InvalidChoiceException & Handle it -At Last use finally to print Thank you for choosing Smart Hub						
		Food Ordering System You are tasked with developing a food ordering system in Java. The system should allow users to select any one food item from a menu, specify the quantity of that item, place their order, and make a payment. Write a Java program that meets the following requirements: 1. Define an abstract class called `FoodItem` with the following attributes and methods: - Attributes: - `name` (protected String). `price` (protected double) - Methods: - `display()`: an abstract method that displays the details of the food item. 2. Define an interface called `Orderable` with the following methods: - `placeOrder(int quantity)`: a method that takes an integer `quantity` as a parameter and only prints a message indicating the order details. - `cancelOrder()`: a method that only prints a cancellation message. 3. Implement two concrete classes, `Pizza` and `Shake`, which inherit from		9				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
113	1,2,3	3. Implement two concrete classes, `Pizza` and `Shake`, which inherit from the `FoodItem` class and implement the `Orderable` interface. Each class should have additional attributes and methods as follows: - `Pizza` class: - Attributes: - `size` (private String): representing the size of the pizza. - Methods: - Implement the abstract `display()` method to display the pizza's details. - `Shake` class: - Attributes: - `size` (private String): representing the size of the shake. - Methods: - Implement the abstract `display()` method to display the shake's details. 4. Define a custom exception class called `InvalidMenuItemException` that extends the `Exception` class. The `InvalidMenuItemException` should have a constructor that accepts a String parameter to display an error message. 5. In the `FoodOrderingSystem` class's `main` method, perform all the following steps: - Create an array of `FoodItem` called `menu` with a capacity of 4. - Initialize the `menu` array with four different food items, including pizzas and shakes, each with different name, price and size as follows:FooditemNamePriceSize PizzaMargherita349Regular PizzaPepperoni449Large ShakeChocolate149Regular ShakeStrawberry249Large - Display the menu by iterating over the `menu` array and calling the `display()` method for each item. - Prompt the user to enter the item number they wish to order. - Validate the user input:- If the item number is invalid (less than 1 or greater than the menu length), throw an `InvalidMenuItemException` with the message "Invalid item number selected!". - If the item number is valid, then prompt the user to enter the quantity to order. - If no exceptions occur, retrieve the selected item from the `menu` array based on the user's input, cast it to `Orderable`, and call the `placeOrder()` method with the specified quantity. - Prompt the user to make a payment by entering a payment method (1 for cash, 2 for credit card). - Implement the `payment()` method with a switch case to handle the payment scenarios: For Cash payment: Show total bill amount and ask user to enter the same						

L.J Institute of Engineering and Technology, Ahmedabad.
Computer Programming using Java-II PRACTICE BOOK (SEM-II-EVEN_2025-CE RELATED BRANCHES)

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
		or more cash_amount. If cash_amount > = bill amount, then payment is successful, display a payment confirmation message. Else throw a custom exception called `PaymentFailedException` with the message "Cash Payment failed!". For Creditcard payment: Show total bill amount and ask user to enter the same or more amount. If credit_amount > =bill amount and Creditcard= 12345678, then only payment is successful, display a payment confirmation message. Else throw a custom exception called `PaymentFailedException` with the message "Creditcard Payment failed!". Handle any exceptions that may occur during the payment process using appropriate exception handling mechanisms. Validation: (You may use below Test cases to validate the functionality of your program before submitting code) 1. Valid item number and quantity: - Select an item from the menu. - Enter a valid quantity. - Make a successful payment. 2. Invalid item number: - Enter an invalid item number (less than 1 or greater than the menu length). - Verify that the program throws the `InvalidMenuItemException`. 3. Payment failure: - Select an item from the menu and Enter a valid quantity. - Choose any payment flow such that it results in a payment failure. - Verify that the program throws the `PaymentFailedException`.						